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Improving Transparency and Reproducibility Through Registration: The Status of Intervention Trials Published in Clinical Psychology Journals

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Objective: Prospective registration increases the validity of randomized controlled trials (RCTs). In the United States, registration is a legal requirement for drugs and devices regulated by the Food and Drug Administration (FDA), and many biomedical journals refuse to publish trials that are not registered. Trials in clinical psychology have not been subject to these requirements; it is unknown to what extent they are registered. ***Method:*** We searched the 25 highest-impact clinical psychology journals that published at least 1 RCT of a health-related psychological intervention in 2013. For included trials, we evaluated their registration status (prospective, retrospective, not registered) and the completeness of their outcome definitions. ***Results:*** We identified 163 articles that reported 165 RCTs; 73 (44%) RCTs were registered, of which only 25 (15%) were registered prospectively. Of registered RCTs, only 42 (58%) indicated their registration status in the publication. Only 2 (1% of all trials) were registered prospectively and defined their primary outcomes completely. For the primary outcome(s), 72 (99%) of all registrations defined the domain, 67 (92%) the time frame, and 48 (66%) the specific measurements. Only 19 (26%) and 5 (7%) defined the specific metric and method of aggregation, respectively, for all primary outcomes. ***Conclusions:*** Very few reports of RCTs published in clinical psychology journals were registered prospectively and completely. Clinical psychology journals could improve transparency and reproducibility, as well as reduce bias, by requiring complete prospective trial registration for publication and by including trial registration numbers in all reports of RCTs.

What is the public health significance of this article?
This study shows that most trials of psychological interventions in clinical psychology journals are not registered in advance, and most registered trials do not define their outcomes completely. The widespread failure to define research hypotheses in advance threatens the validity of trials and meta-analyses, and this has negative impacts on clinical care. Journals should require prospective and complete registration of trials to fulfill ethical obligations to research participants, improve transparency and reproducibility, and reduce publication and outcome reporting bias. All trial reports should state their registration status in the abstract and, if applicable, include their registration number.

Keywords: intervention research, trial registration, randomized controlled trials, reporting guidelines, reporting bias

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Randomized controlled trials (RCTs) are used to evaluate the outcomes of psychological interventions. Despite their advantages, the cumulative body of evidence from RCTs may be invalid if authors, journal editors, or reviewers decide whether to publish based on the magnitude or direction of results (Dickersin, Min, & Meinert, 1992; Emerson et al., 2010; Olson et al., 2002). Withholding studies from publication because of statistically nonsignificant results, or because of results that are contrary to the researcher's hypothesis, is known as *publication bias* (Song et al., 2010), described historically as the "file drawer problem" (Rosenthal, 1979). A related problem, the partial disclosure of information because of the direction or magnitude of results, is known as *selective outcome reporting* (Song et al., 2010). Reporting biases may affect most scientific disciplines (Ioannidis, 2005), including cognitive sciences (Ioannidis, 2012; Ioannidis, Munafò, Fusar-Poli, Nosek, & David, 2014). Reporting bias may be a particular problem in fields like psychology that are dominated by studies with small sample

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sizes (Button et al., 2013), and the prevalence of reporting biases may be worsening over time (Chavalarias, Wallach, Li, & Ioannidis, 2016; Fanelli, 2012). Additionally, effects in psychology may be overestimated because of "p-hacking," the repeated analysis of data following manipulations for the purposes of obtaining results that meet a threshold of statistical significance (Motulsky, 2014); indirect evidence suggests these repeated analyses may be common. For example, a disproportionately larger number of studies report *p* values slightly below 0.05 compared with studies that report *p* values slightly above 0.05 (Chavalarias et al., 2016; Masicampo & Lalande, 2012; Pocock et al., 2004; Simonsohn, Nelson, & Simmons, 2014). Together, these biases increase the likelihood of producing false-positive results (Simmons, Nelson, & Simonsohn, 2011), and they exaggerate the magnitude of true effects (Bakker, van Dijk, & Wicherts, 2012; Ioannidis, 2008).

Studies comparing unpublished and published RCTs in medicine show that RCTs describing treatment outcomes that are statistically nonsignificant are less likely to be published than studies describing statistically significant results (Dwan, Gamble, Williamson, & Kirkham, 2013; Song et al., 2010). Similarly, outcomes in published trials that are statistically nonsignificant may be omitted selectively from published reports (Chan & Altman, 2005; Chan, Hróbjartsson, Haahr, Gøtzsche, & Altman, 2004). Indirect evidence suggests that RCTs in clinical psychology may overestimate intervention effects in the same way (Cuijpers, Smit, Bohlmeijer, Hollon, & Andersson, 2010; Cuijpers, van Straten, Bohlmeijer, Hollon, & Andersson, 2010; Kühberger, Fritz, & Scherndl, 2014; Niemeyer, Musch, & Pietrowsky, 2012, 2013; Staines & Cleland, 2007). Direct evidence also demonstrates that psychological studies submitted to an institutional review board (IRB) with statistically nonsignificant findings are less likely to be submitted for publication compared with studies that have statistically significant findings (Cooper, DeNeve, & Charlton, 1997) and that nonsignificant results in doctoral dissertations are less likely to be published than statistically significant results (Pigott, Valentine, Polanin, Williams, & Canada, 2013). A recent study of trials funded by the National Institute of Mental Health (NIMH) found direct evidence that studies with small effects were less likely to be published than studies with large effects, which may inflate the apparent effectiveness of psychotherapies (Driessen, Hollon, Bockting, Cuijpers, & Turner, 2015).

Underreporting is a long recognized type of scientific misconduct (Chalmers, 1990) that continues to limit the usability of reports of psychological intervention RCTs. In addition to bias, many reports omit information that is essential to understand their conduct and results (Grant, Mayo-Wilson, Melendez-Torres, & Montgomery, 2013) and to synthesize information through systematic reviews and meta analyses (Montgomery, Underhill, Gardner, Operario, & Mayo-Wilson, 2013). As in other disciplines, the nonpublication and partial publication of RCTs in clinical psychology wastes resources and increases the chance that ineffective or harmful interventions will be delivered in practice or tested again in future research (Chan et al., 2014; Glasziou et al., 2014).

For many years, preregistration has been proposed as a solution for improving trial outcome reporting (Dickersin, 1992; Meinert, 1988; Simes, 1986). Trial registers are designed to record important information about study design—especially *complete* descriptions of *all* planned outcome measures. Trial registers provide some guidance for defining outcomes completely and transparently (Zarin, Tse, Williams, Califf, & Ide, 2011). Recording this information in advance, or "prospective registration," allows readers and reviewers to compare the methods and outcomes *reported* in trial manuscripts with the *planned* methods and outcomes. Registrations should be detailed and complete because unclear or ambiguous descriptions of methods and outcomes may allow post hoc manipulation (e.g., p-hacking) or selection based on the results (Mathieu, Boutron, Moher, Altman, & Ravaud, 2009; Zarin et al., 2011).

Registering trials prospectively (i.e., before enrolling participants) is common in medical trials. The Food and Drug Administration Modernization Act of 1997 (FDAMA, 1997) led to the launch of ClinicalTrials.gov in 2000, a publicly accessible database in which RCTs evaluating any type of health intervention can be registered. In 2005, the International Committee of Medical Journal Editors (ICMJE) began requiring prospective trial registration as a condition for publication in its member journals (De Angelis et al., 2004; De Angelis et al., 2005). Hundreds of biomedical journals have adopted this standard, which has also been endorsed by leading organizations including the Council of Science Editors (Council of Science Editors, 2006) and the Committee on Publication Ethics (COPE; Wager & Kleiniert, 2010). Many biomedical journals also require authors to include a trial protocol with their submission. Prospective protocols written using current guidelines (Chan et al., 2013) further increase transparency and reduce the effects of reporting bias; several journals, including *Trials* and *Addiction*, will publish trial protocols. The Food and Drug Administration Amendments Act of 2007 (FDAAA, 2007) requires that all interventional studies of FDA-regulated products (i.e., drugs, biological products, and devices) must be registered. For FDA regulated interventions, trialists must also upload results within 1 year of study completion, though compliance has been poor (Anderson et al., 2015; Marin Dos Santos & Atallah, 2015; Miller, Korn, & Ross, 2015). In addition to these legislative efforts, reporting guidelines emphasize the importance of registration. The Consolidated Standards of Reporting Trials (CONSORT) Statement is an evidence-based guideline for reporting trials that has been endorsed by over 600 journals and editorial groups. CONSORT was revised in 2010 to include "Trial registration number and name of trial registry" as a required item for all RCTs (Schulz, Altman, & Moher, 2010), but use of reporting guidelines in clinical psychology may be limited (Grant et al., 2013).

Clinical psychology is lagging behind efforts in biomedicine to increase research transparency through trial registration, and through *prospective* registration in particular. The American Psychological Association's (APA's) Journal Article Reporting Standards (JARS) acknowledges the possibility of preregistering RCTs, but JARS does not *require* that RCTs be registered (APA Publications and Communications Board Working Group on Journal Article Reporting, 2008). The World Health Organization (WHO) and public campaigns such all AllTrials (alltrials.net) have called for the disclosure of *all* trials, but there is not yet a mechanism to achieve this goal (Moorthy, Karam, Vannice, & Kieny, 2015; World Health Organization, 2015). FDAAA does not apply to psychological interventions; there are no other legal requirements about the registration of social and psychological interventions, and there is not a strong community norm for registration (Faggiano et al., 2014). Although recent reports about transparency

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and data sharing focus on biomedical trials (Institute of Medicine, 2015b), there have been similar calls to apply more rigorous standards to psychological and other nonregulated interventions (Dal-Ré, Bracken, & Ioannidis, 2015; Nosek, Spies, & Motyl, 2012). In the absence of new legal requirements, clinical psychology journals are positioned to play a key role in improving the quality of the research they publish by following ICMJE's example and adopting prospective registration as a requirement for publication.

To investigate the need for a trial registration requirement, we conducted a study to determine the prevalence of registrations and the completeness of outcome definitions in registrations of psychological intervention RCTs published in 25 high-impact clinical psychology journals.

Method

Search Strategy

We selected the 25 journals (a) with the highest impact factor in Thomson Reuters Journal Citation Reports from 2012 (Journal Citation Reports, 2013) in the category "Psychology, Clinical" that also (b) published at least one RCT in 2013. The journal impact factor is the average number of times that articles from a journal published in the preceding 2 years were cited in the index year (i.e., 2012 for this article). Two authors independently searched the table of contents for titles and checked potentially relevant abstracts for all issues of journals in the category "Psychology, Clinical" to identify the first 25 journals publishing at least one RCT and to identify all eligible RCTs of psychological interventions published in those journals. Because reports of RCTs are often not described as RCTs in their titles (Dickersin et al., 2002; Grant et al., 2013), even though this has been recommended in reporting guidelines for almost 20 years (Begg et al., 1996; Schulz et al., 2010), we examined the full-text of articles that referred to an intervention (e.g., Cognitive Behavioral Therapy or "CBT") or contained words such as "treatment," "care," "therapy," or "intervention." We retrieved the full text for any article selected by either author. Because we did not read the full text of every article, some eligible reports with titles or abstracts that appeared to be irrelevant (i.e., those that did not appear to report an intervention study) could have been missed. We resolved discrepancies through discussion and by seeking further information from trialists where necessary.

Two authors reviewed independently the "Instruction to Authors" of included journals to determine whether they referenced reporting guidelines for RCTs and to determine whether they described policies for trial registration. In addition to direct references to reporting guidelines, we also looked for implicit use of the APA reporting guideline, JARS. The APA publication manual incorporates JARS, and we considered a journal to have referenced JARS implicitly if it instructed authors to prepare their manuscript *fully* according to the APA publication manual. We did not consider journals referencing the APA publication manual for preparing specific elements of a publication (e.g., bibliography) to have endorsed JARS either explicitly or implicitly.

Eligibility Criteria

We included RCTs of psychological interventions with human participants. We defined "randomized controlled trial" as (a) "a research study in which one or more human subjects are prospectively assigned to one or more interventions (which may include placebo or other control) to evaluate the effects of those interventions on health-related biomedical or behavioral outcomes" (National Institute of Health, 2014b) and (b) using a random method to assign participants to groups. We did not limit our review by length of intervention; we included single-session interventions and we included trials that measured outcomes immediately following the intervention. We excluded clinical trials in which participants were assigned using nonrandom methods (e.g., alternating assignment). We defined "psychological intervention" as activities, techniques, or strategies that target malleable mental processes and social phenomena (such as perceptions, cognitions, emotions, behaviors, and interpersonal relationships) based on the definition used for the CONSORT Extension for Social and Psychological Interventions (CONSORT-SPI; Grant, 2014), a definition that was adopted in the recent IOM report about psychosocial interventions for mental and substance use disorders (Institute of Medicine, 2015a). Using this definition, we included preventive interventions (which might not include a "clinical" population as psychologists often use the term) as well as studies measuring only proximal outcomes (e.g., intention to change a health-related behavior; Gottfredson et al., 2015). Studies of psychological events that might be associated with future health outcomes are similar to medical studies with "surrogate" endpoints (such as cholesterol or tumor size), and they have similar limitations for predicting patient important outcomes such as functioning and mortality (Kim & Prasad, 2015; Svensson, Menkes, & Lexchin, 2013). Because we sought to evaluate interventions not regulated by FDA, we excluded RCTs in which any participants were randomized to receive a pharmacological intervention. To avoid underestimating the prevalence of registration, and to avoid double counting, we included only the first published report of a trial's results (e.g., the primary report might be more likely than other reports to include registration details). We excluded review articles and other secondary reports of RCTs (e.g., reports describing only mediation analyses or follow-up analyses), which might be less likely to reference trial registrations and more difficult to match with their corresponding trial registrations compared with first reports about the main findings (see Figure 1).

Data Extraction

Two authors independently examined all reports of RCTs for evidence of trial registration (e.g., reference to a registration number). If a relevant report did not include registration information, we contacted authors and searched ClinicalTrials.gov and the International Clinical Trial Registry Platform (ICTRP). For registrations we identified but could not confirm with the authors, we looked for evidence that registrations and publications were related (e.g., by comparing trial location, intervention, number of participants, inclusion/exclusion criteria, and references to publications).

Assessment of Trial Registrations

For registered RCTs, we assessed the extent to which outcomes were defined according to current recommendations (Saldanha,

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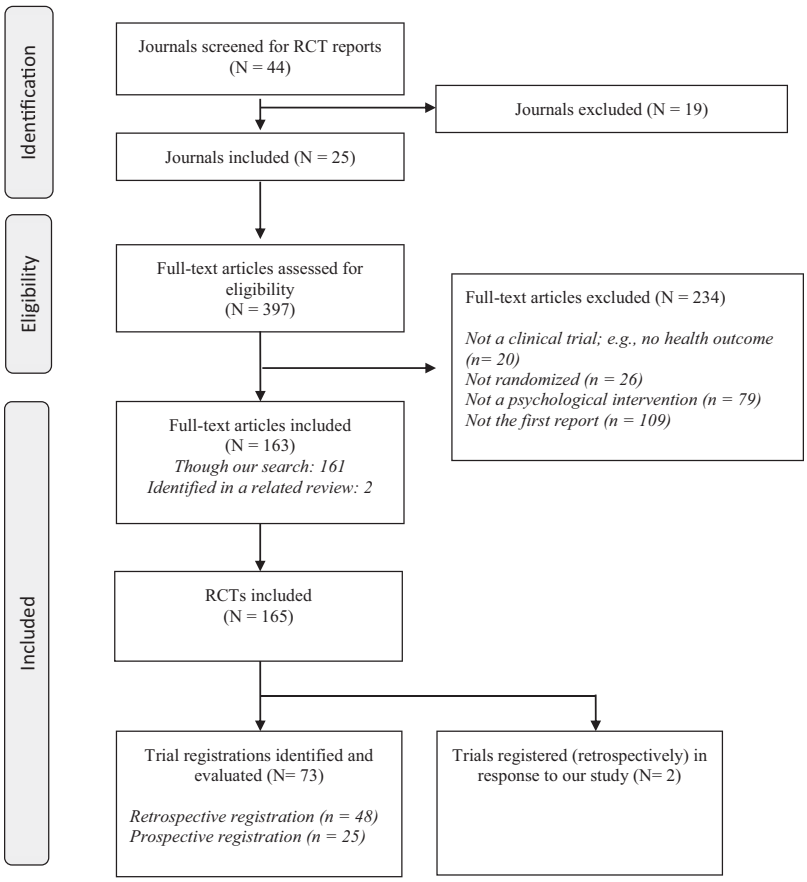


Figure 1. Flowchart showing journal selection, trial selection and proportion of registered trials.

Dickersin, Wang, & Li, 2014; Zarin et al., 2011; see Figure 2), namely: the *domain* (e.g., anxiety), *specific measurement* (e.g., Beck Anxiety Inventory), *specific metric* used to characterize each participant's results (e.g., change from baseline, time to event, final value), the *method for aggregation* of data within each group (e.g., mean, median, proportion of participants), and the time-frame (e.g., baseline, 3 months, 6 months). Two authors independently checked all registrations and recorded if the recommended elements were defined for all outcomes. For prospectively registered trials that were updated after their initial entry, we checked the last version prior to the trial start date; for retrospectively registered trials, we checked the most recent version of the registration. For prospectively registered trials, we also compared the last version prior to the trial start date with the most recent version of the registration. Trial registers all include a section to list study outcomes, but registrations sometimes included information about outcomes outside those sections. For example, NCT00200330 and NCT00255112 listed the time frame for all outcomes in the "detailed description" section. In these cases, we recorded the information as having been reported.

For the primary and secondary outcomes, we recorded whether each element was defined for "all of the outcomes," "some, but not all of the outcomes," or "none of the outcomes." If a registration did not list any outcomes, we considered it to have defined none of the elements (i.e., domain, specific measurement, time frame, etc.) for all primary outcomes. By contrast, we did not code anything about secondary outcomes in registrations that did not list any secondary outcomes because not all RCTs include secondary outcomes. Because an outcome domain may be implicit in the specific measurement (e.g., Beck Depression Inventory), we considered a registered outcome to have defined its domain if a specific measurement with a clear domain was named. We also considered the specific metric to be defined if the registration explicitly stated that the outcome would assess "change." We resolved discrepancies through discussion.

Results

Journal Policies

We searched the 44 journals in the Thomson-Reuters category "Psychology, Clinical" with the highest citation indices to identify the first 25 journals reporting at least one RCT of a psychological intervention; that is, we did not identify any RCTs meeting our inclusion criteria in 19 journals (see Appendix A, in the online supplemental materials). We included the first 25 journals reporting at least one RCT of a psychological intervention. Based on included journals' "Instructions to Authors" (see Appendix B, in the online supplemental materials), we identified references to reporting guidelines in 12 journals (48%; see Appendix C, in the online supplemental materials): One journal (4%) referred to both

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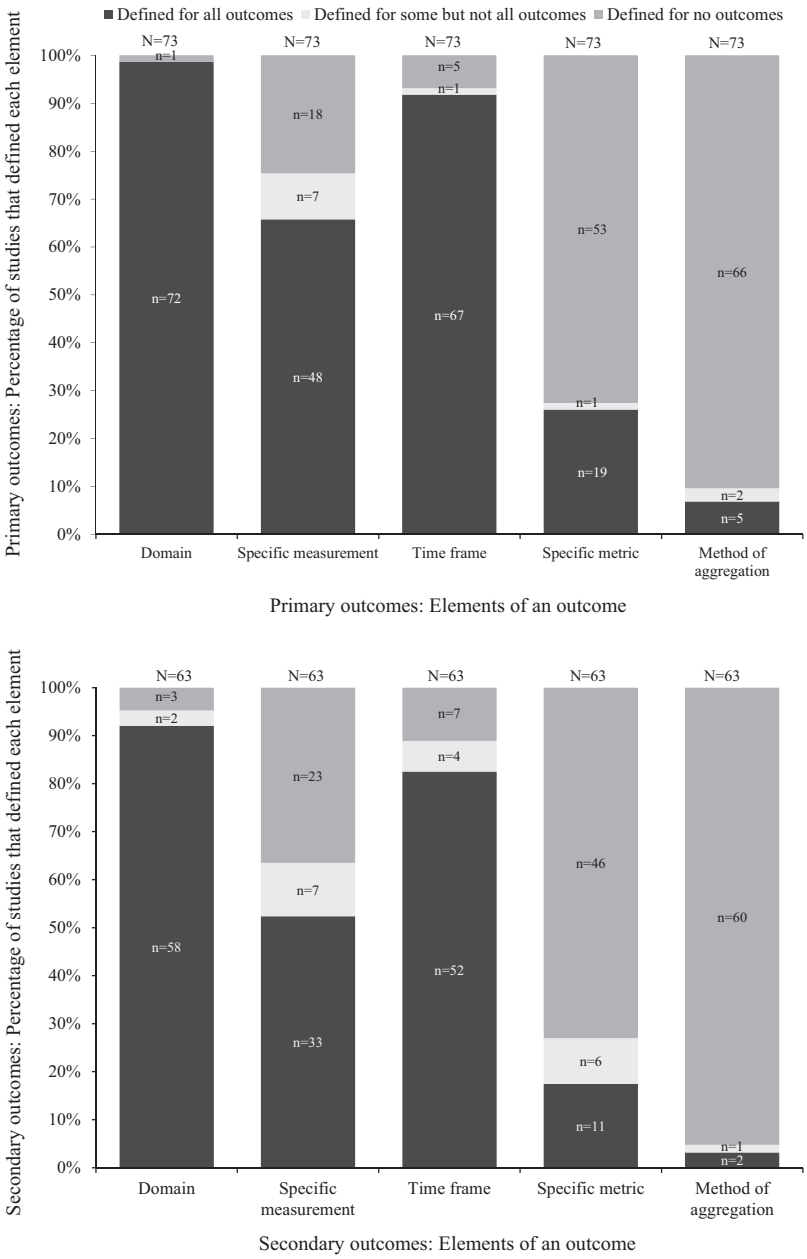


Figure 2. Completeness of trial registrations.

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CONSORT and JARS, nine journals (36%) referred to CONSORT only, and two journals (8%) referred to JARS only (one explicitly and one implicitly by referring to the APA publication manual). Only three included journals stated in their "Instruction to Authors" that it was their policy to publish reports of RCTs only if they were registered prospectively (*Journal of Clinical Psychiatry, Journal of Health Psychology, and International Psychogeriatrics*). In addition, *Journal of Behavior Therapy and Experimental Psychiatry* suggests in its "Instruction to Authors" that "Clinical trials (RCTs and others) should be registered in an official trial register and the registration number should be reported" (see Appendix D, in the online supplemental materials).

Prevalence of Registration

As shown in Figure 1, we identified 397 potentially relevant articles. Of these, 163 (41%) were included and 234 (59%) were excluded; included and excluded studies are described in Appendix E and Appendix F, respectively (see online supplemental materials). Of the 163 included articles, we identified 161 through hand searches. A peer reviewer identified an additional two articles that we missed during our search; these studies were included in a review published after we submitted our article (Azar, Riehm, McKay, & Thombs, 2015). From the 163 included articles, we identified 165 eligible RCTs.

Four journals published 80 (48%) of the 165 RCTs included in this study (see Table 1): *Behavior Research and Therapy* ($n = 28$; 17%), *Journal of Consulting and Clinical Psychology* ($n = 26$; 16%), *Journal of Substance Abuse Treatment* ($n = 15$; 9%), *and Health Psychology* ($n = 14$; 8%).

Of the three journals with a prospective trial registration policy, two of five RCTs in *Journal of Clinical Psychiatry* (40%), none of three RCTs in *Journal of Health Psychology* (0%), and one of six RCTs in *International Psychogeriatrics* (17%) were registered prospectively. One of the three RCTs in *Journal of Clinical Psychiatry* that registered *retrospectively* began in 2000 and was consistent with the journal's policy requiring prospective registration from July 1, 2005. One RCT in *Journal of Health Psychology* registered retrospectively and began in 2006, prior to the journal's policy requiring prospective registration from 2011. Two RCTs in *International Psychogeriatrics* registered retrospectively, neither began before the journal's policy went into effect on December 31, 2006. Neither of the two trials published in *Journal of Behavior Therapy and Experimental Psychiatry* were registered (0%); in one report, the authors remarked, "This study was not registered in the clinical trial register because the authors were uninformed" (Lee & Kwon, 2013).

Of 165 included RCTs, we were able to confirm the registration status for 162 RCTs (98%) because registration numbers were reported in published articles or because we contacted the authors and obtained the registration number (see Table 2). Overall, 73 (44%) RCTs were registered, but only 25 (15%) were registered prospectively (i.e., before or in the same month as the trial start date listed in the registration). Forty-two of 73 (58%) registered RCTs reported the registration number in the published article, including one that reported the registration number in a data supplement. We identified and confirmed with authors an additional 31 (43% of registered trials) registrations that were not referenced in the published article. Of these 31 trials, 29 were registered before we contacted the study authors, of which 8 were registered prospectively, and two were registered by authors response to our enquiries (ACTRN12615000860561 & NCT02517450). We assessed the completeness of registrations for these two trials; the results were consistent with other included studies, but we decided not

Table 1
Characteristics of Included Journals by Impact Factor

Journals	Citation index	Eligible RCTs ($N = 165$)	Registered prospectively ($n = 25$)	Registered retrospectively ($n = 48$)	Trial registration required	Reporting guidance specific to RCTs
Journal of Clinical Psychiatry	5.81	5	2 (40%)	3 (60%)	Yes	No
Psychological Medicine	5.58	6	3 (50%)	2 (34%)	No	No
Journal of Consulting and Clinical Psychology	5.01	26	7 (27%)	6 (23%)	No	JARS
Depression and Anxiety	4.61	5	0 (0%)	4 (80%)	No	No
Health Psychology	3.83	14	2 (14%)	3 (21%)	No	CONSORT
Behaviour Research and Therapy	3.47	28	3 (11%)	7 (25%)	No	No
Psychological Assessment	2.99	1	0 (0%)	0 (0%)	No	No
Behavior Therapy	2.91	9	1 (11%)	4 (44%)	No	CONSORT
International Journal of Eating Disorders	2.87	3	1 (33%)	0 (%)	No	No
Journal of Clinical Child and Adolescent Psychology	2.66	3	1 (33%)	2 (67%)	No	CONSORT; JARS
Journal of Traumatic Stress	2.55	3	1 (33%)	0 (%)	No	CONSORT
International Journal of Behavioral Medicine	2.41	4	0 (0%)	2 (50%)	No	No
Journal of Behavior Therapy and Experimental Psychiatry	2.38	2	0 (0%)	0 (0%)	Recommended	CONSORT
British Journal of Clinical Psychology	2.33	1	0 (0%)	0 (0%)	No	No
Journal of Behavioral Medicine	2.21	5	0 (0%)	1 (20%)	No	CONSORT
International Psychogeriatrics	2.18	6	1 (17%)	2 (33%)	Yes	CONSORT
International Journal of Clinical and Health Psychology	2.10	1	0 (0%)	0 (0%)	No	No
Journal of Positive Behavior Interventions	2.05	1	0 (0%)	0 (0%)	No	JARS[a]
Addictive Behaviors	2.02	10	1 (10%)	2 (20%)	No	No
Clinical Psychology and Psychotherapy	2.00	6	0 (0%)	1 (17%)	No	No
British Journal of Health Psychology	1.99	5	1 (20%)	0 (0%)	No	CONSORT
Journal of Substance Abuse Treatment	1.98	15	1 (7%)	7 (47%)	No	No
European Eating Disorders Review	1.94	1	0 (0%)	1 (100%)	No	No
Journal of Family Psychology	1.89	2	0 (0%)	0 (0%)	No	CONSORT
Journal of Health Psychology	1.88	3	0 (0%)	1 (33%)	Yes	CONSORT

Note. Citation index = Thomson-Reuter citation index for 2012, 1-year impact factor; Eligible RCTs = Number of RCTs that met the inclusion criteria published in 2013; Registered prospectively = RCTs registered prior to the start date in a trial register (*e.g.*, clinicaltrials.gov); Registered retrospectively = RCTs registered after the start date. Trial registration required: Yes = Journal "Instruction to Authors" states that publication is conditional on prospective preregistration; No = Journal "Instruction to Authors" does not state that publication is conditional on prospective registration; Recommended = Journal "Instruction to Authors" does not state that publication is conditional on prospective registration, but that it is recommended. Reporting guidelines specific to RCTs = Reporting guidelines mentioned in the "Instruction to Authors" section of the journal web site.
[a] These journals referred to JARS implicitly by referring to the APA publication manual.

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Table 2
Status of Included Studies

Registration status	Included studies (N = 165)	Registered prospectively (N = 25)	Registered retrospectively (N = 50)
Not registered, confirmed with author	89 (54%)	—	—
Not registered, unconfirmed	1 (1%)	—	—
Registered and reported in publication	42 (25%)	17 (63%)	25 (52%)
Registered, confirmed with author	29 (18%)	8 (30%)	21 (44%)
Registered, unconfirmed	2 (1%)	0 (0%)	2 (4%)
Registered in response to our enquiry (retrospective)	2 (1%)	0 (0%)	2 (7%)

Note. Not registered = Corresponding author confirmed that the trial was not registered; Not found = We could not identify a matching registration by contacting the authors or by searching trial registers; Registered, unconfirmed = Registration identified by searching clinicaltrials.gov or ICTRN, but not confirmed by the author; Registered prospectively = RCTs registered prior to the start date in a trial register (*e.g.*, clinicaltrials.gov); Registered retrospectively = RCTs registered after the start date.

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to include registrations completed in response to our enquiry as part of this report (see Appendix G, in the online supplemental materials).

By contacting authors, we confirmed that 89 of 90 (99%) RCTs for which we did not locate a registration were not registered. In response to our request for information about the registration status of their trials, some authors gave reasons for not registering, including "It was conducted very early in my career and I did not yet know to register it;" "It was a brief experimental study that was not funded by any national-level grants;" and "[T]his study was a pilot study of a new intervention that we were developing, to investigate acceptability and feasibility for a full-scale clinical trial." We were unable to confirm trial registration status of three RCTs (2%); we identified registrations that we believe are associated with two of those trials (Bergman, Gonzalez, Piacentini, & Keller, 2013; Ironson et al., 2013), and we believe the third was not registered (Stice, Rohde, Durant, Shaw, & Wade, 2013).

Completeness of Registration

Of 73 registered trials, 63 (86%) of the trial registrations included at least one primary outcome and at least one secondary outcome; eight (11%) registrations defined only primary outcomes and no secondary outcomes; one (1%) registration did not did not define any primary and secondary outcomes; and one (1%) registration listed outcomes but did not specify which were primary or secondary (we coded all the listed outcomes as primary; see Appendix G, in the online supplemental materials). We did not compare prospective and retrospective registrations because our goal was to describe the completeness of outcome definitions.

Domain, specific measurement, and time frame were registered commonly for primary outcomes. Seventy-two (99%) registrations defined the outcome domain (i.e., defined explicitly or implicitly in the specific measurement listed) for all primary outcomes. Time frame was defined for all primary outcomes in 67 (92%) registrations, and the specific measurement was defined for all primary outcomes in 48 (66%) registrations. All three of the above elements were defined for all primary outcomes in 48 (66%) registrations. Nineteen (26%) registrations defined the specific metric for all primary outcomes. The method of aggregation was defined for all primary outcomes in five (7%) registrations (see Figure 2). Only four trials (5% of all registered RCTs, 2% of all included RCTs) registered all outcomes completely (i.e., defined all outcome measure elements for all the listed primary outcomes). Examples of registered primary outcomes are included in Table 3.

Sixty-three (86%) registrations defined secondary outcomes. The domain was defined for all secondary outcomes in 58 (92%) registrations. The time frame was defined for all secondary outcomes in 52 (83%) registrations, and the specific measurement was defined for all secondary outcomes in 33 (52%) registrations. Domain, specific measurement, and time frame were registered for all secondary outcomes in 32 (51% of registrations with secondary outcomes) registrations. Only 11 (17%) registrations defined the metrics for all secondary outcomes, and two (3%) registrations defined the method of aggregation for the secondary outcomes (see Figure 2).

Discussion

Main Results

Our finding that only 15% of trials were registered prospectively is worrisome for the transparency, reproducibility, and potential reporting biases in the evidence underpinning psychological interventions. On the other hand, our finding that almost half of identified trials were registered (prospectively or retrospectively) is somewhat encouraging because most clinical psychology journals and funders do not require that researchers register psychological intervention trials. Moreover, two trialists registered their studies in response to our enquiries, which is promising for further research about interventions to inform trialists about registration and to encourage prospective registration. These results are consistent with recent evidence that the rate of registration and the use of reporting guidelines in psychiatry journals is relatively low (Knuppel, Metz, Meerpohl, & Strech, 2013; Scott, Rucklidge, & Mulder, 2015) and evidence that trial registration has increased in four leading behavioral health journals from 20% to 53%, coinciding with three of those journals adopting policies that mandate trial registration (Milette, Roseman, & Thombs, 2011; Riehm, Azar, & Thombs, 2015). Our overall results are also similar to the results of a review focused on one journal, which concluded that most studies are not registered and that many registrations

Table 3
Examples of Registered Primary Outcomes

Elements included (trial registration number)	Description of primary outcome	Elements we recorded as having been defined in the registration
Domain (NTR1352)	"Behavior problems"	Domain: behavior problems Specific measurement: not defined Specific metric: not defined Method of aggregation: not defined Time frame: not defined
Domain, specific measurement, and time frame (NCT00218959)	"Clinician Administered PTSD Scale (Time frame: before treatment, 1 month after, 6 months after)"	Domain: posttraumatic stress Specific measurement: Clinician Administered PTSD Scale (CAPS) Specific metric: not defined Method of aggregation: not defined Time frame: before treatment, 1, and 6 months after
Domain, specific measurement, time frame, and specific metric (NCT00626457)	"Change in body weight (Time frame: 12-month follow-up period [6–18 months])" "Body weight will be measured at baseline, 6, 12, and 18 months on a California state-certified standard beam balance scale at the research center with participants in light clothing and without shoes"	Domain: body weight Specific measurement: California State-Certified Standard Beam Balance Scale Specific metric: Change[a] Method of aggregation: not defined Time frame: at baseline, 6, 12, and 18 months
Domain, specific measurement, time frame, and method of aggregation (NCT00484302)	"Number of cannabis-abstinent days (Time frame: 1 month) . . . Assessment of number of days with cannabis abuse during the last month with Time Line Follow-Back" (Sobell and Sobell, 1992) "Continuous outcome measures will be analyzed with ANOVA analyses of variance. Dropout analysis will be carried out comparing the group of patients who complete the entire experimental treatment with the population who enrolled in the trial. Baseline values of measures are included as covariates in the analysis whenever possible. To take the dropout into account repeated measurement (mixed model) analysis with interaction analysis of time and type of treatment is used, in order to evaluate the effect over time."	Domain: abstinence Specific measurement: number of cannabis-abstinent days in past month, measured with time line follow-back Specific metric: change from baseline[b] Method of aggregation: continuous, comparison of means Time frame: at baseline, 6, and 10 months[c]

Note. We did not include an example of a registered primary outcome that defined the elements domain and specific measurement only because we did not find an example including only these elements.
[a] We assumed change from baseline. [b] The metric is not included in the "Primary outcome" field. We inferred that the metric is "change from baseline" using two pieces of information. First, "baseline" is reported under "Effect measurements" in the "Detailed description" section. Second, the "Detailed description" section describes "Statistics," including an ANOVA to test for between-group differences in change over time. [c] Although we coded the time frame as having been reported completely, it is not clear if 6 and 10 months refer to time since baseline, time since randomization, time since the end of treatment, or time since some other starting point.

are incomplete (Azar et al., 2015). Furthermore, the completeness of outcome definitions in some registrations indicates that it is possible for psychological intervention studies to include all recommended outcome elements. Given that none of the trial registers are structured to prompt and require that authors describe each element of an outcome, it was also encouraging that some authors chose to provide this information voluntarily.

Despite evidence that registration is possible and increasing among trials of psychological interventions published in leading journals, it is scientifically and ethically concerning that most registrations were *retrospective*, many reports did not indicate trial registration status, and registrations were overwhelmingly incomplete. Specifically, registrations commonly omitted information about the metric and method of aggregation. Some of the trials registered prospectively were registered before the publication of guidance for outcome registration (Zarin et al., 2011), which is the standard we used for assessing completeness. In some cases, trialists updated those registrations to meet the new requirements retrospectively and we would have rated their registrations as more complete if we had used the most recent version (see Appendix H, in the online supplemental materials).

The relatively high number of retrospective registrations may reflect growing understanding of the importance of trial registration and an attempt to correct existing deficiencies. Retrospective registration is preferable to nonregistration because registration can help researchers identify previous trials and because it can help systematic reviewers link multiple reports about trials (Altman, Furberg, Grimshaw, & Shanahan, 2014). Prospective registration is needed for all *future* trials (i.e., those not yet completed or recruiting participants) because retrospective registrations are insufficient for identifying and preventing reporting biases. Retrospective registrations cannot show if intended research plans were followed and if all planned outcomes were reported; some evidence suggests that trialists do not reliably report the outcomes they measured (Chan, Hróbjartsson, et al., 2004; Chan, Krleza-Jeric, Schmid, & Altman, 2004). Our study did not seek to address the prevalence of reporting bias or the adequacy of statistical analysis

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plans. Changes in outcomes between registrations, changes in outcomes between registration and publication, and changes between any preanalysis plans and reported analysis plans are important topics but did not necessarily affect our rating of completeness of outcome definitions.

Changing Social Norms

Authors reported lack of awareness as a reason for nonregistration, and they also reported differences in opinion about what constitutes an "intervention" and a "trial" that should be registered. This feedback from authors could be useful in raising awareness and changing social norms about prospective registration of interventions trials by psychology researchers. For example, one author responded "This was not a trial but rather an experimental study that included a very brief intervention of about 20–30 minutes. It was thus not registered as a clinical trial." Another said "[B]ased on criteria on the government web site looks like our trial would be excluded" (we included this randomized evaluation of a CBT-based stress management and resilience training program because it met all of the criteria for this review, and the authors are correct that their study was not subject to mandatory registration requirements). We included a study measuring "intentions to undertake recommended levels of physical activity" that the authors replied lacked "a behavioral or health outcome (so therefore consistent with the NIH definition). I would just call it an experiment." As these examples show, it can be difficult to distinguish what many psychologists describe as "experiments" from what NIH calls "clinical trials." As author feedback demonstrates, we included several randomized evaluations that other authors might have excluded.

Author feedback also suggests that some investigators believe that "pilot," "feasibility," or "acceptability" trials do not need to be registered, yet these are the studies that may be most vulnerable to reporting bias. For instance, one author replied "I cannot recall if it was eligible to be registered since it was a pilot—if it is eligible I am happy to do so." While understanding trial phase may be important for interpreting study findings (Saitz, 2014), *all* randomized trials should be registered prospectively so that readers can interpret them correctly. Tests for publication bias detect *small study bias*, which occurs because very large definitive studies are relatively likely to be published regardless of their results; on the other hand, small studies with statistically significant results are much more likely to be published than small studies with statistically nonsignificant results. Registering small, early phase trials is essential to avoid the overrepresentation of small studies with statistically significant results in the scientific literature.

Editors play a clear role in the prospective registration of trials, the reporting of registration status, and the completeness of outcome definitions in trial registrations. For example, one author replied that

> I had included the registry and number in the abstract of my final manuscript submission . . . but [the journal] removed it when they sent back the page proofs. When I posted the final version of the submitted manuscript to PubMed Central, which is available to the public, I made sure the registry and number stayed in the abstract.

One editor replied "I don't know for certain whether this study was registered but it would seem unlikely to me as it is not a clinical trial." The editor of a journal that requires trial registration explained that one trial was not registered and that "the journal's requirement for trial registration was waived in the case of this particular submission as we viewed the study as an experiment, rather than a more traditional clinical trial." One study reported the registration number in a data supplement, which is useful for the interested reader; however, an electronic search for reports related to the registration number might miss that report. Journals could increase transparency by requiring the reporting of registration status for all randomized trials and by including this information in the abstract and article.

Strengths and Limitations

We report a comprehensive survey of 25 influential journals in clinical psychology. There is little risk of underestimating the prevalence of registration among included studies. By contacting authors, we confirmed the registration status of almost all relevant reports, including all but one study that we described as "not registered." A limitation of this review is that we might have missed some eligible trials because they did not describe clearly the intervention and study design in the title or abstract (Grant et al., 2013). A reviewer identified two reports that we missed; the title "Converging to the Lowest Common Denominator in Physical Health" did not suggest anything about an intervention or a trial, and we did not include "Monitoring of Exhaled Carbon Monoxide to Promote Preoperative Smoking Abstinence" because the title suggests that the intervention was an ineligible physical exam (in fact, both groups were examined; the independent variable was *informing* participants about monitoring). Missing a few eligible studies would not change our conclusions. Studies that do not adhere to CONSORT or JARS guidelines for titles and abstracts might be less likely to be registered compared with studies that comply with reporting guidelines, so our review may describe a best-case scenario. Our results are consistent with previous reviews focused on specific psychology journals and psychiatry journals, and they are consistent with evidence that trial registrations in general are incomplete (Viergever, Karam, Reis, & Ghersi, 2014; Zarin et al., 2011).

The results of our study were influenced by several decisions. For example, our definition of a "psychological intervention" was purposefully broad, which led us to include several RCTs that some authors argue do not need to be registered. In coding registrations, we considered information about an outcome to be present if it appeared in any section (e.g., time frame appearing only in the title). We also judged the domain to have been reported if a questionnaire was named. We interpreted "change" as a sufficient description of metric. We recorded that a specific measurement was reported even if it could be analyzed in multiple ways. (, e.g., "SF-36" might be considered incomplete because the questionnaire is not typically analyzed using a total score; it would be better to describe which components or subscales will be analyzed, such as "SF-36, Mental Health Component.") For these reasons, our report may overestimate registration prevalence and completeness in high impact clinical psychology journals.

We did not review all journals in in the Thomson-Reuters citation index category "Psychology, Clinical." Most RCTs were published in high impact journals in this category (which have similar impact and relative ranking from year-to-year); it is un-

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likely that including other journals in the category would change the overall conclusions. We also did not include the many journals that publish relevant trials but do not appear in the category "Psychology, Clinical," including journals in psychiatry (e.g., *American Journal of Psychiatry*), public health (e.g., *American Journal of Public Health*), and medicine (e.g., *British Medical Journal*). Many public health journals and most medical journals endorse CONSORT guidelines and require prospective trial registration (http://www.consort-statement.org/), so RCTs published in those journals might be more completely registered and more completely reported than those published in the journals we reviewed.

A few trials published in 2013 began before trial registration was possible. For example, one author replied that the trial "was launched in 1998 when registration was not practiced." Other studies were initiated before current guidance was available, so authors might not have known how to define their outcomes completely (Zarin et al., 2011), though we found examples of nearly complete registrations before these guidelines were written (e.g., NCT00484302 was registered in 2007, see Table 3). We did not record the date of article submission, and we do not know when journals adopted current guidelines; some journals might have adopted trial registration policies or started to endorse reporting guidelines after some trials were submitted for publication. For these reasons, we hope to see continued improvement in the prevalence and completeness of trial registration as a result of changes already underway. To evaluate the effects of these changes, future studies could compare changes over time and evaluate the correspondence of registered outcomes and published outcomes.

To answer our question about completeness of outcome definitions, we followed guidance from Zarin, Tse, Williams, Califf, and Ide (2011), who distinguish between outcome definitions and methods of analysis. Thus, we considered the number of outcomes and the statistical methods to be beyond our scope. Some authors have suggested that trial registries should require that authors either include a single primary outcome or describe plans for statistical adjustment because multiplicity may contribute to Type I errors (Azar et al., 2015; Riehm et al., 2015). However, methods to adjust for multiple outcomes reflect the enduring use of *p* values, and there is widespread scientific consensus that tests for "automatic application of dichotomized hypothesis testing based on prearranged levels of statistical significance" (Kyriacou, 2016) are insufficient for interpretation. To estimate the size of intervention effects and uncertainty in those estimates, trials should report tests that describe precision, such as confidence intervals (Wasserstein & Lazar, 2016). Furthermore, multiple outcomes may *not* be associated with bias if they are defined in full, registered prospectively, and reported completely using appropriate tests for magnitude and uncertainty.

Efforts to Improve Transparency

Sustained efforts to promote registration are important because reporting biases undermine the validity of systematic reviews and clinical guidelines, negatively affect people seeking psychological therapies (i.e., by increasing use of interventions that are less beneficial than published reports suggest), and waste scarce resources (Chan et al., 2014; Glasziou et al., 2014; Macleod et al., 2014). To advance knowledge and to help people with similar problems, participants in clinical trials volunteer to receive untested treatments, and they may experience delays in accessing services. For these reasons, it is ethically incumbent upon trialists to share the findings of their research (Antes & Chalmers, 2003; Chalmers, 1990; Scherer, Langenberg, & von Elm, 2007). Funders, academic institutions, and the public are increasingly involved in actions to redress research bias and waste (Moher et al., 2015). In particular, editors of medical journals helped lead the campaign to make trial registration the norm in their discipline, and editors of clinical psychology journals could do the same.

Existing trial registers provide the infrastructure needed to register studies of health interventions, including psychological interventions. Adhering to existing guidance for registering outcomes would improve transparency, and current recommendations could be improved by instructing trialists to describe statistical analysis plans (e.g., "preanalysis plan;" Miguel et al., 2014), including planned adjustments for multiple outcomes and covariates and plans for handling missing data. Several ongoing initiatives aim to increase awareness about reporting bias and to improve research transparency. There have been multiple calls to increase transparency in social science research (APA Data Sharing Working Group, 2015; Gonzales & Cunningham, 2015; Miguel et al., 2014; Wagenmakers, Wetzels, Borsboom, van der Maas, & Kievit, 2012; van Assen, van Aert, Nuijten, & Wicherts, 2014). Evidence of outright fraud (Oransky, 2015) and the failure to replicate many influential psychological experiments (Open Science Collaboration, 2015) have drawn national attention to the importance of good research conduct and transparent reporting. The Berkeley Initiative for Transparency in the Social Sciences (http://bitss.org), The Data Access and Research Transparency Statement (http://dartstatement.org), and the Center for Open Science (http://centerforopenscience.org) are all contributing to standards and platforms that foster research integrity and transparency in research conduct and dissemination. Despite these prominent efforts, most clinical psychology journals continue to lag behind biomedicine in adopting fundamental transparency standards.

The culture of publishing reports based on the results may be changing, albeit slowly. At the time of writing, 19 journals have agreed to accept studies for publication *before* they collect and report results on the condition that the studies are methodologically sound and scientifically important (see Center for Open Science, 2015b for a regularly updated list), and the Center for Open Science has introduced a set of "badges" that journals can award researchers to acknowledge and incentivize transparency in research conduct (Center for Open Science, 2015a). Several journals, including *Perspectives on Psychological Science* (Simons, Holcombe, & Spellman, 2014) and *Psychological Science* (Eich, 2014), have published editorials announcing these new editorial policies. Knowing a study will be published *regardless of the results* may remove one incentive for manipulation or omission, though other concerns including funding and promotion may continue to encourage reporting biases. Perhaps most importantly, the National Institutes of Health (NIH) recently proposed to require that *all* NIH-funded clinical trials be preregistered and post their results (National Institute of Health, 2014a). If NIH adopts a rule requiring prospective registration and posting of summary results, responses to our enquiries suggest it may be useful for NIH to provide guidance about which behavioral and psychological studies the institutes consider "clinical trials." The authors and editors we contacted used the term "clinical trial" in different ways, and they used other terms to describe "clinical trials." For example, they often described studies about health behaviors outside of clinical settings as "experiments." There is growing consensus that *all* studies in which

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human participants are prospectively assigned to one or more interventions should be registered regardless of phase, setting, nature of the intervention (manipulation), and type of outcomes (Goldacre, 2015).

Reporting guidelines can also help authors and journal editors increase research transparency. The Transparency and Openness Promotion (TOP) Guidelines provide examples of good editorial policies and include instructions to authors that journals can adopt with minimal effort (Nosek et al., 2015). Reporting guidelines for trials of psychological interventions, such as the CONSORT Extension for Social and Psychological Interventions (CONSORT-SPI), can help authors write more complete and transparent reports (Mayo-Wilson et al., 2013; Montgomery, Grant et al., 2013). Specific guidance also exists for particular types of psychological interventions such as group-based behavior-change interventions (Borek, Abraham, Smith, Greaves, & Tarrant, 2015), and the EQUATOR Group provides toolkits to help authors and editors identify and implement reporting guidelines (Simera et al., 2010). Adherence to reporting guidelines would complement efforts to promote ethical standards and research transparency through trial registration.

Educational efforts could address some perceived obstacles to registration. Specifically, registration is an uncomplicated process that can be completed quickly online (Zarin & Keselman, 2007). In 2014, ClinicalTrials.gov launched a training site (National Library of Medicine, 2015b) with expanded guidelines to help trialists register their studies (Zarin, 2014). Trial registration is free, is not limited to FDA regulated products, and is available for both funded and unfunded studies. The registration of studies is also not an obstacle to exploratory analyses (Wagenmakers et al., 2012). Finally, the term "clinical trial" does not imply that participants must have a health problem; studies to develop, test, or evaluate any intervention (that is, activities intended to improve functioning, promote health behaviors, prevent disease, or treat health issues) are all eligible for registration (National Institute of Health, 2014b; National Library of Medicine, 2015a).

Journal enforcement of prospective trial registration is a crucial step to meeting our ethical obligations and to preventing multiple analyses and reporting biases that can distort the evidence base for psychological interventions (Gorman, 2015). Policies *mandating* trial preregistration are necessary because encouragement alone does not appear to produce desired change (Thaler et al., 2015) while active implementation is associated with increased transparency (Pandis, Shamseer, Kokich, Fleming, & Moher, 2014). Journals that publish RCTs of psychological interventions could increase transparency and improve trust in psychological interventions by publishing only those RCTs that have been registered prospectively, as required for ICMJE journals. Enforcing such policies could reduce reporting bias, allow researchers to demonstrate fidelity in trial conduct and reporting (e.g., by comparing process evaluations with trial registrations; Moore et al., 2015), increase the visibility of psychological research by making records available through a public database, improve clinical outcomes by identifying the interventions most likely to maximize benefit and reduce harm (Kaplan & Irvin, 2015), and help funders allocate resources more effectively (i.e., by reducing research waste). Journals could announce new requirements in editorials, such as one published in *International Psychogeriatrics* (Ames, 2006). Several organizations provide suggestions and templates for author guidelines, including EQUATOR (EQUATOR Network, 2015). We encourage journals to endorse registration by adding the following policy to their Instructions to Authors and by adopting internal procedures that enforce this policy:

> [Journal] will publish reports of randomized controlled trials (RCTs) only if they have been registered (i.e., in an independent, publicly accessible register). To be considered for publication, all trials commencing recruitment after [date] must be registered prospectively (i.e., before commencing recruitment). Trials commencing recruitment before [same date as above] must be registered no later than [same date as above or the date of submission, whichever is earlier]. A full list of acceptable registers can be found via the WHO International Clinical Trials Registry Platform (ICTRP; http://www.who.int/ictrp/network/trds/en/). Registrations must include all elements of each primary and secondary outcome (Zarin et al., 2011), including the times at which each outcome will be measured and analyzed. The name of the trial register and the trial registration number should appear on the front page of the manuscript and at the end of the abstract. All differences between (a) the reported methods and outcomes and (b) the registered methods and outcomes must be described in the manuscript and explained in a cover letter.

Journals that wish to adopt other best practices to promote transparency could also add the following:

> The trial protocol must be submitted as a supplementary file and include recommended items (Chan et al., 2013). The protocol will be provided to peer reviewers and published as an online supplement if the manuscript is accepted for publication.

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